

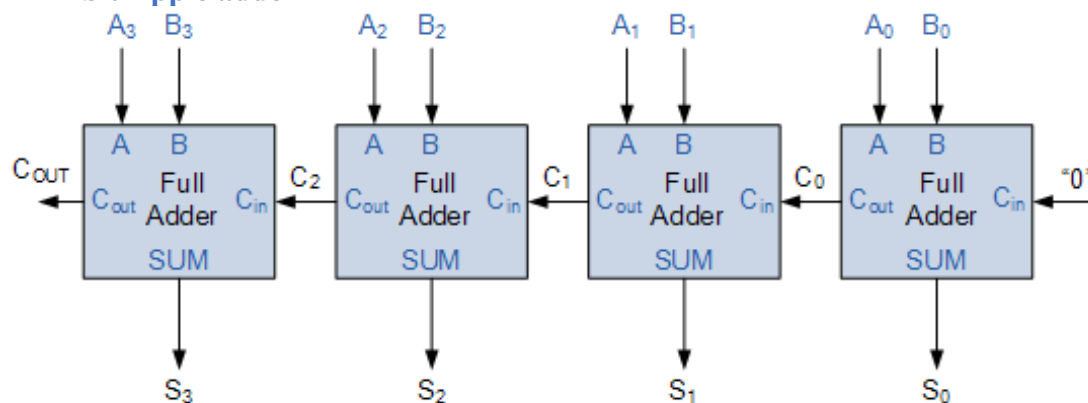
2024 Digital Intergrated Circuits HW4

Using 7nm FinFET devices with $V_{DD} = 0.8V$, FF process corner and medium V_t CMOS process. Rise time and fall time of input signals and clock are 0.02ns (0V 0.8V) 7nm tech. path: /RAID2/COURSE/dic/dicTA01/DIC_2024_Fall/7nm_files

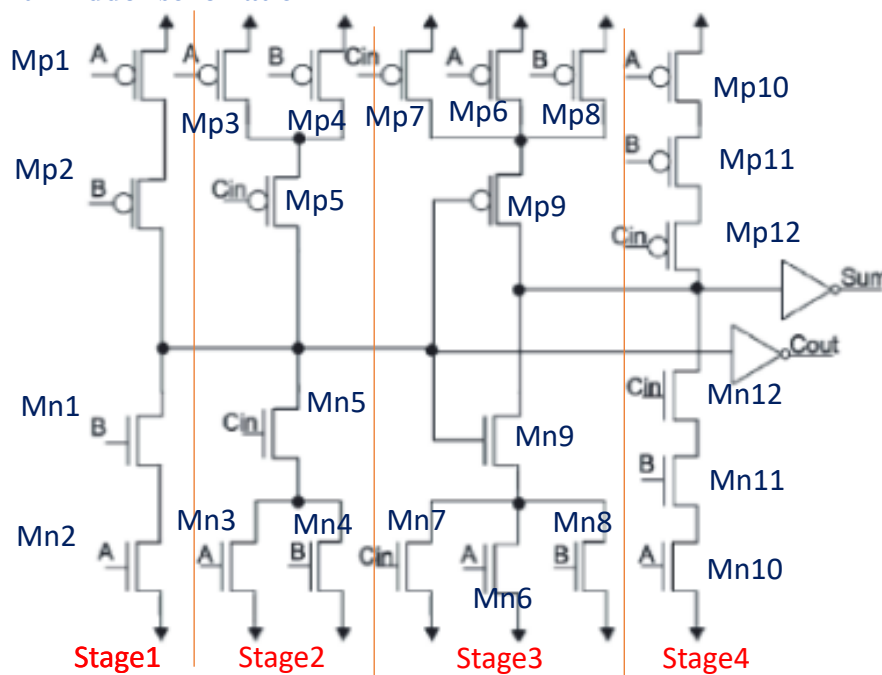
A 4-bit ripple carry adder shown in Fig. 1 is designed with fully complementary static logic gate of full adder (FA). Input signals are $A[3:0]$, $B[3:0]$, and C_{in} . Outputs are $SUM[3:0]$ and C_{out} .

- (1) Try to design the fastest adder when each input signal is driven by one unit size inverter. For the output loads, 5 unit size inverters are used for each output signal. First, show your block diagrams in terms of the 1-bit FA. Second, show the circuit schematic of the FA. Use logic effort concepts (do not have to write down the procedure) to design **transistor widths**. Describe **your design concept**. (40%)

4bit ripple adder



Full Adder schematic



Resource: https://www.researchgate.net/publication/340807074_CMOS_Based_1-Bit_Full_Adder_Cell_for_Low-Power_Delay_Product

從 1-bit FA 的方塊圖及其電路示意圖可以觀察到，關鍵路徑必定是從 Cin 到 S3，而在這過程中，每個 FA 的 Cout 都會持續傳遞至下一級的 Cin。根據邏輯設計原理，我們知道只要能平衡每一級的驅動能力，就能有效降低傳遞延遲。因此，透過增強並平衡 Cout 的驅動能力，即可達到降低 tpd 的目的。由於 Cout 和 Cin 的驅動能力主要與 FA 電路示意圖中的 stage2 有關（這是因為需要等待上一級的 Cout 傳遞到當前的 Cin），所以我們只需要著重於加強並平衡每一級 stage2 的驅動能力，就能有效降低 tpd。

(2) Based on the design of (1), run SPICE to find the propagation delay (As shown in Fig. 2, with pattern from $IA[3:0] = 4'b1111$, $IB[3:0] = 4'b0000$, $ICin = 1'b1$ to $IA[3:0] = 4'b1111$, $IB[3:0] = 4'b0000$, $ICin = 1'b0$). Determine the maximum propagation delay of the Ripple Carry Adder (exclude the load inverters). (20%)

以下為與 unit size inveter size 相同推動力的 FA size 以及他的 propagation delay:
Mp1~Mp12 nfin=1

```
***** transient analysis tnom= 25.000 temp= 25.000 *****
max_propagation_delay= 56.9544p targ= 2.0734n trig= 2.0165n
```

以下增加 stage2 推動力的 FA size 以及他的 propagation delay:
Mp3~Mp5 nfin=2, Mp3~Mp5 nfin=2

```
***** transient analysis tnom= 25.000 temp= 25.000 *****
max_propagation_delay= 49.3293p targ= 2.0670n trig= 2.0176n
```

以下為經過多次實驗所得到最好的 propagation delay 以及 transistor size:

finfet	nfin
Stage1(Mp1,Mp2,Mn1,Mn2)	1
Stage2(Mp3,Mp4,Mn4,Mn5)	6
Stage2(Mp5, Mn3)	3
Stage3(Mp6,Mp7,Mp8,Mp9,Mn6,Mn7,Mn8,Mn9)	1
Stage4(Mp10,Mp11,Mp12,Mn10,Mn11,Mn12)	1

```
***** transient analysis tnom= 25.000 temp= 25.000 *****
max_propagation_delay= 38.9669p targ= 2.0577n trig= 2.0188n
```

(3) Run SPICE of the ripple carry adder with each output load equals to 4 unit size inverter. Determine the **average, peak, and leakage** power dissipation and energy per bit, respectively when simulating at the working frequency in (2). (20%)

依照下方 pattern 以及第二小提最後計算最佳的 propagation delay 作為 working frequency 的依據，即 $\frac{1}{40ps} = 25GHz$

Table I. Input Pattern for (3)

Average and Peak Power	Leakage Power
A[3:0] = 0000, 0001, ... 1111	A[3:0] = 0000
B[3:0] = 1111, 1110, ... 0000	B[3:0] = 1111
C _{in} is 0/1 by turns	C _{in} = 0

```
.tran 0.001ps 680ps

.measure tran power_avg avg P(xripple)
.measure tran power_peak max P(xripple)

.measure tran energy_per_bit param = 'power_avg*0.0000000068/4'

.measure tran leakage_power avg P(xripple) FROM=0p TO=40p
```

```
***** transient analysis tnom= 25.000 temp= 25.000 *****
power_avg= 108.1748u from= 0. to= 680.0000p
power_peak= 300.0737u at= 338.5026p
           from= 0. to= 680.0000p
energy_per_bit= 18.3897f
leakage_power= 2.0135n from= 0. to= 40.0000p
```

Power_avg	108.1748u Watt
Power_peak	300.0737u Watt
Energy_per_bit	18.3897f J
Leakage_power	2.0135n Watt

(4) Add pipelining stages as shown in Fig. 4 into the 4-bit ripple carry adder, with the D register given in Fig. 3. Run SPICE to find the propagation delay time (with pattern from $IA[3:0] = 4'b0000$, $IB[3:0] = 4'b1111$, $ICin = 1'b0$ to $IA[3:0] = 4'b0000$, $IB[3:0] = 4'b1111$, $ICin = 1'b1$) between pipelining stages to determine the maximum working frequency of the clock.

算電路的 propagation delay，在下次 clk 來之前必須要將值運算完畢，因此最快的工作頻率為 period 的倒數。

```
.measure tran period trig v(clk) val=0.4 rise=4
+               targ v(sum3) val=0.4 fall=1

.measure tran max_working_freq param='1/period'
.end
```

```
***** transient analysis tnom= 25.000 temp= 25.000 *****
period= 45.2318p  targ= 286.2318p  trig= 241.0000p
max_working_freq= 22.1084g
```

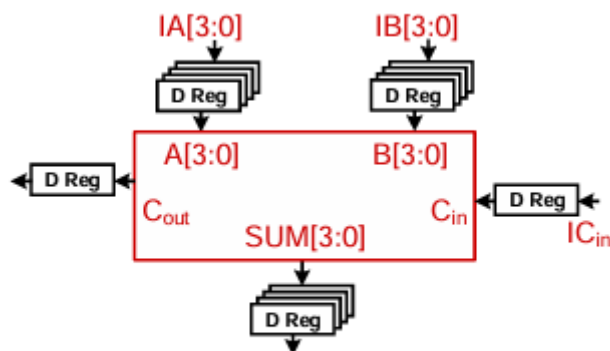


Fig. 4.